The primitive rank-two geometries of M_{11}

This note treats Tables 3 and 4 of Leemans and Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. 2 (2019), P1.08. The fixed journal edition has twelve pages. Its numerical suborbit table and its total of 37 unordered subgroup-pair orbits were already recovered by the common character and mark calculation. Here the remaining task is to recover the individual graph invariants and design parameters from concrete subgroup pairs. The earlier calculation is an independent comparison, not an input prescribing the number of pairs.

1 The finite input and its scope

We use the acquired degree-eleven model

$$G = \langle (1\,2\,\cdots\,11),\ (3\,7\,11\,8)(4\,10\,5\,6) \rangle$$

and its retained array of 7920 elements. The accepted numerical subgroup census and structural proofs used for L054 supply 39 subgroup-class representatives. Completeness of that imported census is an explicit premise. Its five maximal classes, in their retained order, are

class	38	37	35	34	27
order	720	660	144	120	48
index	11	12	55	66	165.

The names M_{10}, $L_2(11)$, $M_9 : 2$, S_5 and $Q_8 : S_3$ are imported from the actual accepted structural proofs for these literal representatives. The last name is identified with $\mathrm{GL}_2(3)$ below. We also retain the classical fact $\mathrm{Out}(M_{11}) = 1$. Thus inner conjugacy of unordered pairs is exactly the equivalence defined by the source. No restriction such as the intersection property $(\mathrm{IP})_2$ is imposed.

Lemma 1. *Let G be a finite group with a complete list of maximal subgroup classes, and let Ω be the union of their conjugacy orbits. If a supplied partition of $\binom{\Omega}{2}$ is closed under generators of G, and each part is connected under those generator actions, its parts are exactly the conjugacy orbits of unordered pairs of distinct maximal subgroups.*

Proof. Closure under a generator implies closure under its inverse, since the part is finite and the generator acts injectively. Thus each part is G-invariant and is a union of orbits. Connectivity means that every member can be reached from the chosen representative by generators and inverses, so that the part lies in one orbit. The two inclusions give equality. □

The certificate retains every maximal subgroup as an actual subset of G, every conjugator from its class representative, the two generator actions, and the complete partition of the unordered-pair domain. Consequently completeness is checked on a finite domain, rather than inferred from agreement with a printed total.

2 Cosets, distances and cycles

For a retained pair H, K, construct its full right-coset partitions. The edge set is

$$\{(Hg, Kg) : g \in G\}.$$

These are exactly the pairs of cosets with nonempty intersection: an element in the intersection supplies g, and conversely the displayed two cosets both contain g. An edge is represented by exactly $|H \cap K|$ elements, so there are $|G|/|H \cap K|$ edges. Right multiplication is transitive on each vertex type and on the edges. Our permutations use this right-action convention, exactly as the source does. If a prior component uses left cosets, inversion supplies the precise bridge $Hg \mapsto g^{-1}H$: two right cosets intersect if and only if their inverse left cosets intersect. It preserves types and distances, while right multiplication by a becomes left multiplication by a^{-1}. No exchange of the two types is part of this bridge.

Distinct maximal subgroups generate G. The incidence graph is therefore connected. Its two degrees are $[H : H \cap K]$ and $[K : H \cap K]$, both at least two, so it is firm. The action on each type is primitive by maximality. Thus every retained pair gives a geometry in the source's class. All these assertions also have literal finite checks in the certificate.

The acquisition computes all vertex eccentricities and the global girth. The independent verifier uses one vertex of each type for the eccentricity and a different shortest-cycle calculation. For a chosen root r, delete r, initialize a separate search region at each neighbor of r, and run a simultaneous breadth-first search. A meeting of two regions gives a shortest path between two different neighbors. Its length plus two is the length of a shortest cycle through r. Conversely every cycle through r gives such a path after deleting r. Each type is transitive under the explicit right action of G, and any shortest cycle contains both types. Hence each chosen root lies on a cycle of global minimum length. This justifies the rooted verification; a single arbitrary rooted breadth-first collision would not, without further argument, certify global girth. The gonality is half the girth, as in the source.

For the intersection name, the certificate gives a literal $g \in G$ and a retained subgroup class L such that $H \cap K = g^{-1}Lg$. Equality is checked on complete element sets. Only then is the accepted structural proof for L imported. Equal subgroup orders are never used as a structural test. When the source prints an integer alone, the recovered assertion is only that intersection order.

3 The order-forty-eight name

The acquired representative of class 27 is generated by

$$a = (1\,2\,9\,4\,6\,5\,3\,10)(7\,11), \qquad b = (2\,3)(4\,10)(5\,9)(8\,11).$$

A complete finite isomorphism sends these generators to

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix} \quad \text{over } \mathbf{F}_3.$$

The certificate lists one matrix image for each of the 48 actual permutations. The verifier checks that the images are precisely all invertible 2 by 2 matrices, that identity maps to identity, and that $f(xy) = f(x)f(y)$ for all $48^2 = 2304$ pairs. This is a bijective homomorphism, and proves the literal $\mathrm{GL}_2(3)$ identification.

For comparison with the accepted extension structure, put

$$i = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad j = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad r = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad s = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Here $\langle i, j\rangle \cong Q_8$, with $i^2 = j^2 = -I$ and $j^{-1}ij = i^{-1}$, while $\langle r, s\rangle \cong S_3$, since $r^3 = s^2 = 1$ and $srs = r^{-1}$. Their intersection is trivial and their products exhaust the 48 invertible matrices. The actual retained S_3 complement acts faithfully on its normal Q_8, as is verified by checking its centralizer on all eight elements. The explicit full isomorphism above supplies the identification even without invoking a classification of actions on Q_8.

4 Designs and orientations

For either choice of point type, the blocks are the neighborhoods of the vertices of the other type. The certificate retains their complete point sets, their constant size k, their number, and the number of distinct neighborhoods. For each claimed strength t, the verifier counts, for every t-subset T of points, the blocks containing T. A constant value λ proves precisely the stated $S_\lambda(t, k, v)$ condition. The first failed strength is certified by two explicit subsets with unequal counts.

No larger nonvacuous strength can hold after such a failure. Indeed, for a fixed t-subset T, count pairs (x, B) with $x \notin T$ and $T \cup \{x\} \subseteq B$. If every $(t+1)$-subset has multiplicity λ_{t+1}, this gives

$$(k - t)\,\#\{B : T \subseteq B\} = (v - t)\lambda_{t+1},$$

so the multiplicity at level t is constant whenever $t < k$.

The point type in the design table is identified by its actual point count, block size and multiplicity. The join records this orientation separately from the ordering of the two subgroup names in Table 3. If the printed invariants leave two rows indistinguishable, all permissible matches are retained and a bijection is selected only within that ambiguity. No unique embedding is inferred from an incomplete list of invariants. Complete incidence is recorded separately: every block is the full point set, and every subset is contained in all blocks. Whether such an object is called a design depends on whether the convention requires proper or distinct blocks. The literal counts make that convention visible.

5 The source comparison and four orientation corrections

The actual computation gives 309 maximal subgroups and partitions all $\binom{309}{2} = 47586$ unordered pairs into 37 orbits. The independent checker validates 6278 literal cosets and 28498 subset counts in the uniform design levels. Every asserted intersection name is joined through its actual conjugator to an accepted structural proof. The extra matrix certificate identifies the order-48 maximal subgroup.

Keeping the subgroup columns printed in Table 3, four triples require correction:

source row	printed	corrected
12, 18	[3, 2, 4]	[4, 2, 3]
23, 35	[3, 3, 4]	[4, 3, 3].

Section 3 of the source defines d_p as the eccentricity of a G_1 coset and d_l as that of a G_2 coset. The four printed triples instead use the opposite orientation. This is a correction of the stated convention; it does not change the abstract graphs. The original transcription and its failed literal comparison are retained separately from the explicit correction layer. All 70 numerical design entries in Table 4 agree.

Row 35 has a short independent interpretation. Its point sets in the opposite orientation have sizes 11 and 55, and the design is the full set of two-element subsets of eleven points. Its incidence graph is the subdivision of K_{11}. An original vertex has maximum distance three, attained at a nonincident edge vertex. An edge vertex has maximum distance four, attained

at a disjoint edge vertex. The source puts the index-55 subgroup $M_9 : 2$ first and the index-11 subgroup M_{10} second, so its own convention gives $[4, 3, 3]$. Its gonality is three because a triangle of K_{11} becomes a cycle of length six. The same argument gives the row 23 correction with K_{12}.

Rows 12 and 18 are likewise recovered with the stated type sizes. In the orientation with 12 or 11 points, their point diameters are three and block diameters four; the subgroup columns in Table 3 put the blocks first. The full finite distance certificates establish both values. Table 4's design orientation is retained independently of these subgroup columns.

Rows 10 and 11, and rows 16 and 17, have indistinguishable printed profiles. The computation supplies two different pair orbits in each case. We retain both possible source matches within each pair of rows, rather than pretend that the printed invariants identify a unique embedding. This ambiguity does not change any asserted table entry or the complete orbit census.

The omitted design rows 31 and 32 are respectively complete bipartite graphs on parts of sizes $(12, 55)$ and $(12, 11)$. In each orientation every block is the full point set. They satisfy the literal subset-counting condition with repeated full blocks; their omission uses the usual requirement of proper or distinct blocks, which is not stated in the source's short definition. We record both the counts and this convention qualification. No nontrivial design entry is lost.

6 Attribution and boundaries

The subgroup classification, its named structural results and the standard properties of M_{11} are borrowed with exact links to their accepted components. The source supplies the assertions to be compared. Our work here is the common finite certificate, its independent verification, the explicit source joins and the matrix naming bridge. None of these finite checks is presented as Lean verification or as a new compression measurement. The complete source disposition requires the returned graph data and source comparison to pass independent review.

7 Recovered finite tables

The subgroup columns follow the source. A dagger marks an explicitly corrected orientation; the two type diameters have been exchanged while the subgroup columns are retained. Bare integers in the intersection column assert orders only.

Row	G_1	G_2	$G_1 \cap G_2$	$[d_p, g, d_l]$
1	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	S_3	$[5,3,5]$
2	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	C_2^2	$[4,2,4]$
3	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	C_2	$[4,2,4]$
4	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	C_2	$[5,2,5]$
5	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	C_2	$[3,2,4]$
6	$\mathrm{GL}_2(3)$	$\mathrm{GL}_2(3)$	1	$[3,2,3]$
7	$\mathrm{GL}_2(3)$	S_5	D_{12}	$[5,3,5]$
8	$\mathrm{GL}_2(3)$	S_5	D_8	$[4,2,4]$
9	$\mathrm{GL}_2(3)$	S_5	S_3	$[4,2,4]$
10	$\mathrm{GL}_2(3)$	S_5	C_2	$[3,2,3]$
11	$\mathrm{GL}_2(3)$	S_5	C_2	$[3,2,3]$
$12^\dagger$	$\mathrm{GL}_2(3)$	$L_2(11)$	D_{12}	$[4,2,3]$
13	$\mathrm{GL}_2(3)$	$L_2(11)$	S_3	$[3,2,3]$
14	$\mathrm{GL}_2(3)$	$M_9:2$	$Q_8:2$	$[6,3,5]$
15	$\mathrm{GL}_2(3)$	$M_9:2$	D_{12}	$[4,3,4]$
16	$\mathrm{GL}_2(3)$	$M_9:2$	C_2	$[3,2,3]$
17	$\mathrm{GL}_2(3)$	$M_9:2$	C_2	$[3,2,3]$
$18^\dagger$	$\mathrm{GL}_2(3)$	M_{10}	$Q_8:2$	$[4,2,3]$
19	$\mathrm{GL}_2(3)$	M_{10}	S_3	$[3,2,3]$
20	S_5	S_5	D_8	$[5,2,5]$
21	S_5	S_5	S_3	$[3,2,3]$
22	S_5	S_5	C_2^2	$[3,2,3]$
$23^\dagger$	S_5	$L_2(11)$	A_5	$[4,3,3]$
24	S_5	$L_2(11)$	12	$[3,2,3]$
25	S_5	$M_9:2$	12	$[4,2,3]$
26	S_5	$M_9:2$	8	$[3,2,3]$
27	S_5	$M_9:2$	C_4	$[3,2,3]$
28	S_5	M_{10}	24	$[3,2,3]$
29	S_5	M_{10}	20	$[3,2,3]$
30	$L_2(11)$	$L_2(11)$	A_5	$[3,2,3]$
31	$L_2(11)$	$M_9:2$	D_{12}	$[2,2,2]$
32	$L_2(11)$	M_{10}	A_5	$[2,2,2]$
33	$M_9:2$	$M_9:2$	8	$[3,2,3]$
34	$M_9:2$	$M_9:2$	4	$[3,2,3]$
$35^\dagger$	$M_9:2$	M_{10}	M_9	$[4,3,3]$
36	$M_9:2$	M_{10}	16	$[3,2,3]$
37	M_{10}	M_{10}	M_9	$[3,2,3]$

All seventy printed design parameters are recovered without numerical changes. The two design columns use the point orientation of source Table 4; the exact join records where this reverses the subgroup ordering above. Rows 31 and 32 are omitted in the source and are complete incidence, as discussed in the text.

Row	Source design	Source dual design
1	$S_8(1,8,165)$	$S_8(1,8,165)$
2	$S_{12}(1,12,165)$	$S_{12}(1,12,165)$
3	$S_{24}(1,24,165)$	$S_{24}(1,24,165)$
4	$S_{24}(1,24,165)$	$S_{24}(1,24,165)$
5	$S_{24}(1,24,165)$	$S_{24}(1,24,165)$
6	$S_{48}(1,48,165)$	$S_{48}(1,48,165)$
7	$S_{10}(1,4,66)$	$S_4(1,10,165)$
8	$S_{15}(1,6,66)$	$S_6(1,15,165)$
9	$S_{20}(1,8,66)$	$S_8(1,20,165)$
10	$S_{60}(1,24,66)$	$S_{24}(1,60,165)$
11	$S_{60}(1,24,66)$	$S_{24}(1,60,165)$
12	$S_3(3,4,12)$	$S_4(1,55,165)$
13	$S_{42}(3,8,12)$	$S_8(1,110,165)$
14	$S_3(1,9,165)$	$S_9(1,3,55)$
15	$S_4(1,12,165)$	$S_{12}(1,4,55)$
16	$S_{24}(1,72,165)$	$S_{72}(1,24,55)$
17	$S_{24}(1,72,165)$	$S_{72}(1,24,55)$
18	$S_1(3,3,11)$	$S_3(1,45,165)$
19	$S_1(8,8,11)$	$S_8(1,120,165)$
20	$S_{15}(1,15,66)$	$S_{15}(1,15,66)$
21	$S_{20}(1,20,66)$	$S_{20}(1,20,66)$
22	$S_{30}(1,30,66)$	$S_{30}(1,30,66)$
23	$S_1(2,2,12)$	$S_2(1,11,66)$
24	$S_1(10,10,12)$	$S_{10}(1,55,66)$
25	$S_{10}(1,12,66)$	$S_{12}(1,10,55)$
26	$S_{15}(1,18,66)$	$S_{18}(1,15,55)$
27	$S_{30}(1,36,66)$	$S_{36}(1,30,55)$
28	$S_5(1,30,66)$	$S_1(4,5,11)$
29	$S_6(1,36,66)$	$S_3(4,6,11)$
30	$S_1(11,11,12)$	$S_1(11,11,12)$
33	$S_{18}(1,18,55)$	$S_{18}(1,18,55)$
34	$S_{36}(1,36,55)$	$S_{36}(1,36,55)$
35	$S_1(2,2,11)$	$S_2(1,10,55)$
36	$S_1(9,9,11)$	$S_9(1,45,55)$
37	$S_1(10,10,11)$	$S_1(10,10,11)$